

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent, in the Rocq prover. The first semantics closely models the ELPI’s implementation, it is based on a stack of choice points and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on and-or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics (1) to prove properties about the effect of the cut operator in the evaluation of choice points, since the rules from classical logic do not apply, and (2) to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [18, 19, 8, 13]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [14, 3, 4, 25, 9, 10]. Other remarkable libraries include the Hierarchy-Builder library-structuring tool [5] and Derive [23, 24, 12], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [11] to help users control backtracking. ROCQ users are familiar with functional programming, but not necessarily with logic programming; in particular, uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker allows the user to assert which predicates should be considered deterministic and then checks that these predicates behave like functions, i.e., they commit to their first solution and leave no *choice points* (places where backtracking could resume). A choice point correspond to an suspended *alternative* in the execution trace. In the following we use *choice point* and *alternative* as synonyms.

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [11]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [20] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations



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```

func f int -> int. % f is a deterministic predicate. The two rules
f 1 2.             % have non-verlapping heads, i.e. 1 != 2
f 2 3.
pred r int -> int. % r is a non-deterministic predicate. The heads
r 0 0.             % overlap: the query for 0 has three applicable
r 0 1.             % rules.
r 0 2 :- !.
func g int -> int. % g is deterministic: if the first rule succeeds, the
g A C :- r A B, f B C, !. % second one is cut away, as well as the choice
g D F :- f D E, f E F.   % points left by the call to r.

```

■ **Figure 1** Running example of a ELPI program

used by standard PROLOG abstract machines [26, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by cut are explicit and we prove the two semantics equivalent. Using the tree-based semantics, we then prove some properties about the impact of evaluating the cut in disjunctive queries. For example, unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 .

Finally, we use the formal semantics to prove the correctness of a static determinacy analysis [11]. Intuitively, nondeterminism arises when a computation can be completed by applying two different rules. We exclude this situation in two ways. Firstly, at most one rule should be used on a given query. This condition is satisfied if (i) the rules have non-overlapping heads: if two heads do not overlap in the inputs they accept, then no query can unify with both, or if (ii) we account for the presence of cut in rule premises: once a rule whose premises contain a cut succeeds, all remaining rules are discarded. Secondly, each rule of a deterministic predicate should (i) either call functions, this ensure that no choice point is left after the execution of the premises of the chosen rule, (ii) or call relations provided that they are followed by the cut operator, so that any allocated choice points preceeding the cut are discarded.

We show that if every rule defining a deterministic predicate passes this analysis, then any call to that predicate leaves no choice points.

2 Preliminaries: syntax and informal semantics of cut

In the entire paper we use Figure 1 as our running example, and we start by describing how we represent a program like that one. In the concrete syntax, variables are written with capital letters, and predicate application follows the λ -calculus style (no parentheses).

The description of a program is given by the record $\mathbb{P}$ and is shared by both semantics. A program consists of a list of rules R together with a signature environment sigT . We delay the discussion of signatures to Section 6 where we describe the static analysis that concerns them.

```

Inductive Tm := Symb of S | Pred of P | Var of V | App of Tm & Tm.
Inductive Atom := cut | call of Tm.
Record R := mkR { head : Tm; premises : list Atom }.
Record P := { rules : seq R; sig : sigT }.
Definition Σ := {fmap V -> Tm}.

```

■ **Figure 2** Definition of term, rule, program and substitution

A rule consists of a head term Tm and a list of premises. Each premise is an Atom , i.e. either the cut operator or a predicate call. Terms Tm include predicate symbols, data symbols, and variables. The syntax tree allows variables to be applied and predicates to be mixed with data. These higher order features play no role in the current paper that focuses on first order logic programming, but allows future extensions to full λProlog to be based on the same mechanization.

The set of variables $\mathbf{V}$, predicates $\mathbf{P}$ and data symbols $\mathbf{S}$ are countable. We represent substitutions Σ as finitely supported maps from variables to terms, using the `finmap` library [17]. The empty substitution is written ε while the composition of two substitutions σ_1 and σ_2 with disjoint supports as $\sigma_1 + \sigma_2$.

The backchaining operation applies all rules to a query term; it is central to both semantics.

Definition `bc` : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \text{Tm} \rightarrow \Sigma \rightarrow \mathcal{F}_v * \text{list } (\Sigma * \text{list } \text{Atom})$

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the `backchain` function takes a program, the set of variable names used so far ($\mathcal{F}_v$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term.

In our mechanization, we abstract over the unifier and its properties. The function `unify` takes two terms t_1 and t_2 , together with a substitution σ , as input, and returns an optional substitution σ' that extends σ . The expected behavior of `unify` is higher-order unification restricted to the pattern fragment, as explained in [18]. Therefore, if t_1 and t_2 unify under σ , then $\sigma' t_1 \equiv \sigma' t_2$, where σt is the notation for substitution application and $\equiv$ denotes the syntactic equality of its arguments.

2.1 The cut operator

There are quite a few variants of the cut operator. The one we are interested in, i.e. the one used in ELPI, is known as *hard cut* (see, e.g., the documentation of SWI-PROLOG [22]). Whenever the `backchain` function returns a non-empty list of alternatives, the first one is explored immediately, while the others are suspended as *choice points*. Within a given alternative, premises are processed from left to right, executing each atom in turn. When a cut is encountered, it discards (i) all choice points created by `backchain` for the current call and (ii) all choice points created while executing the premises to the left of the cut.

For example, consider the program in Figure 1 and the query `g 0 R`. Both rules for `g` apply to this query. Backchaining therefore returns the following two alternatives:

$[(\sigma_1, [\text{r } \mathbf{A} \ \mathbf{B}, \ \mathbf{f} \ \mathbf{B} \ \mathbf{C}, \ !]), (\sigma_2, [\mathbf{f} \ \mathbf{D} \ \mathbf{E}, \ \mathbf{f} \ \mathbf{E} \ \mathbf{F}])]$

with $\sigma_1 := \{\mathbf{A} \mapsto 0, \mathbf{C} \mapsto \mathbf{R}\}$ and $\sigma_2 := \{\mathbf{D} \mapsto 0, \mathbf{F} \mapsto \mathbf{R}\}$. For simplicity, we choose an example where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_v$ any further.

Execution continues with the first premise of the first alternative, namely `r A B` under σ_1 , i.e. `r 0 B`. The remaining alternatives are stored as choice points. Backchaining `r 0 B` returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [!])]$ where $\sigma_3 := \sigma_1 + \{\mathbf{B} \mapsto 0\}$; $\sigma_4 := \sigma_1 + \{\mathbf{B} \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{\mathbf{B} \mapsto 2\}$.

The next step of interpretation continues with `f B C` under σ_3 , that is, `f 0 R`, and leaves $[(\sigma_4, []), (\sigma_5, [!])]$ as new choice points. This time, backchaining fails (no rule for `f` matches input 0). We therefore backtrack to the most recently introduced choice point and retry `f B C` under σ_4 , i.e. `f 1 R`. This succeeds with $\sigma_6 := \sigma_4 + \{\mathbf{R} \mapsto 2\}$.

usiamo + ?

```

Inductive tree :=
  | KO | OK                                (* failure and success *)
  | Todo of Atom                          (* unexplored branch *)
  | Or  of option tree &  $\Sigma$  & tree      (*  $A \vee B_\sigma$  *)
  | And of tree & list Atom & tree.       (*  $A \wedge_{B_0} B$  *)

```

■ **Figure 3** Definition of And-Or tree

We now reach the cut. At this point there are still two unexplored alternatives: one from the third rule for `r` and one from the second rule for `g` (respectively, σ_5 and σ_2). Both are discarded, because they were created when, or after, the first rule for `g` was selected. (In contrast, a cut does not discard choice points created earlier; in this example, there are none.)

The final solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$.

In Sections 3 and 4, we provide fully mechanized operational semantics for PROLOG with cut. They differ in how they represent the ongoing computation, in particular how alternatives are stored and how they are pruned by cut.

Notational conventions In the following section we note $\mathbb{B}$ for the boolean type, and $\top$ and $\perp$ for `true` and `false`. The constructors of the option type are written $\cdot t$ for `Some t` and $\square$ for `None`. Following the Mathematical Components library, on which we depend, we use `x.1` and `x.2` to denote the first and second projection applied to the pair `x`.

3 And-Or Tree semantics

An And-Or tree is a stratified, variable-width tree. It consists of two kinds of layers that alternate: a disjunctive layer is followed by a conjunctive layer and vice versa. At the root (a query), the tree records a list of alternatives (an Or-layer); for each alternative, it records a list of subqueries to be solved (an And-layer). Intuitively, solving a query requires solving one alternative in the Or-layer, but all subqueries in the corresponding And-layer.

The tree is built incrementally. The `Todo` node represents an unexplored branch (an `Atom`). When the atom is a `call`, we use the backchaining procedure from Section 2 to compute the two layers to attach to the tree: alternatives form the Or-layer, and the premises of each applicable rule form the corresponding And-layer. This expansion step is performed by the `step` procedure described in Section 3.2.

The Rocq data type for And-Or trees is given in Section 3. In addition to `Todo`, it provides two distinguished nodes, `KO` and `OK`, which represent respectively the smallest failed and successful tree.

The `Or` node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing disjunction A . The left subtree A is optional and is absent once that branch has been fully explored. The right alternative is paired with a substitution σ , which becomes the current substitution when backtracking to B .

The `And` node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to the first choice point in A . We call B_0 the *reset point* for B : it represents the continuation for all alternatives of A except the first. That is, when backtracking occurs within A , the subtree B is replaced by the atoms in B_0 . An example is shown in Section 3.3.

A graphical representation of a tree is shown in Figure 6b. For compactness, we depict `And` and `Or` nodes as n -ary (rather than binary), with children associating to the right. The

spiegare nota-
tion for list

TODO: nested
option, output
Fv also inside op-
tion
corner case
prog2.elpi

why option and
not KO?

```

Inductive tag := Expanded | CutBrothers | Failed | Success.
Definition step :  $\mathbb{P} \rightarrow \mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F} * \text{tag} * \text{tree})$ 
Definition prune :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree}$ 
Inductive runT :  $\mathbb{P} \rightarrow \mathcal{F} \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F} \rightarrow \text{Prop}$ 

```

■ **Figure 4** Signatures of the tree-based semantics

155 KO node acts as the terminating element of an Or list, while OK is the terminating element of
 156 an And list.

157 3.1 The tree-based semantics

158 The tree is constructed incrementally by two recursive functions, **step** and **prune**. Both
 159 traverse the tree depth-first, left-to-right. The node to be explored in a tree is retrieved
 160 thanks to the **next_tree** (in Figure 5). When this node is OK we say the entire tree is a
 161 *success*, when it is KO we say the tree *failed*, otherwise the tree is *incomplete*.

162 Since all functions must terminate in Rocq, we define their transitive closure of **step**
 163 and **prune** as the inductive relation **runT**. More precisely **runT** iterates calls to **step** on
 164 incomplete trees to expand **Todo** nodes. When a tree fails it calls **prune** to find the next
 165 alternative and continues from there, if one exists. We detail **step**, **prune**, and **runT** in
 166 Sections 3.2–3.4.

167 In Figure 6 we mark with a black arrow the result of **next_tree**.

168 3.2 The step procedure

169 The **step** procedure takes as input a program, a set of variables, a substitution, and a tree,
 170 and returns an updated set of variables, a **tag**, and an updated tree.

171 The set of variables is assumed to contain all variables occurring in the tree. It is passed
 172 to backchain so that rules can be refreshed correctly.

173 The **tag** (see Section 3.1) indicates which kind of step was performed. **Failure** and **Success**
 174 are returned for trees that satisfy, respectively, the **failed** and **success** tests. **CutBrothers**
 175 indicates the interpretation of a superficial cut: **step** was called on an And-layer and its

¹ The substitutions $\sigma_1 \dots \sigma_6$ are from Section 2.1. R_1, R_2 , and R_3 are reset points and correspond respectively to the lists of atoms $[f \ Y \ Z, !]$, $[!]$, and $[f \ Y \ Z]$.

```

Fixpoint next  $\sigma \ t : \Sigma * \text{tree} :=$ 
  match  $t$  with
  | Todo _ | KO | OK  $\Rightarrow (\sigma, t)$ 
  | Or  $\square \ \sigma' \ B \Rightarrow \text{next } \sigma' \ B$ 
  | Or  $\cdot A \ \_ \Rightarrow \text{next } \sigma \ A$ 
  | And  $A \ \_ \ B \Rightarrow$ 
    let  $(\sigma', pA) := \text{next } \sigma \ A$  in
    if  $pA == \text{OK}$  then  $\text{next } \sigma' \ B$ 
    else  $(\sigma', pA)$ 
  end.

Definition next_subst  $\sigma \ t := (\text{next } \sigma \ t).1.$ 
Definition next_tree  $t := (\text{next } \varepsilon \ t).2.$ 

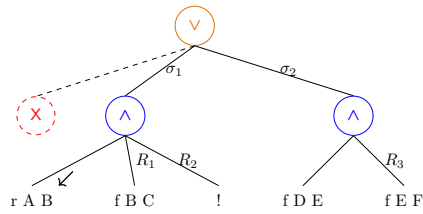
Definition success  $t := \text{next\_tree } t == \text{OK}.$ 
Definition failed  $t := \text{next\_tree } t == \text{KO}.$ 
Definition incomplete  $t :=$ 
  if  $\text{next\_tree } t$  is Todo _ then  $\top$  else  $\perp.$ 

```

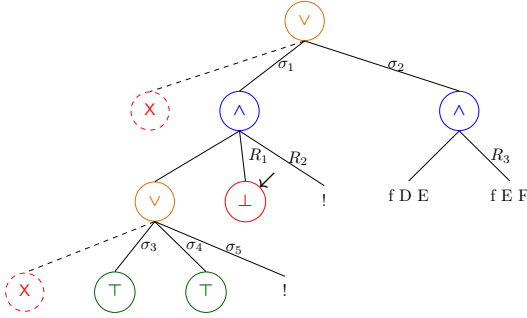
■ **Figure 5** next and derived functions

g 0 R

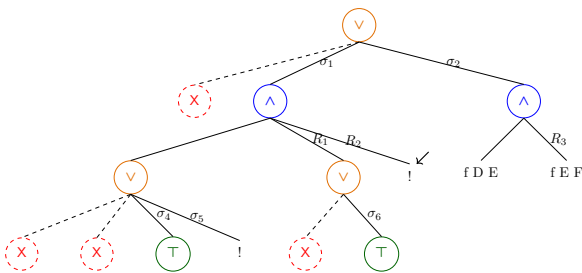
(a) Initial query



(b) Tree after step on g 0 R

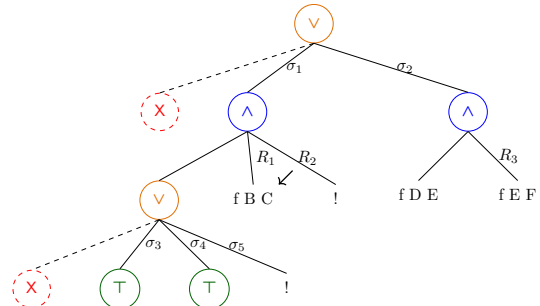


(d.1) step on Figure 6c

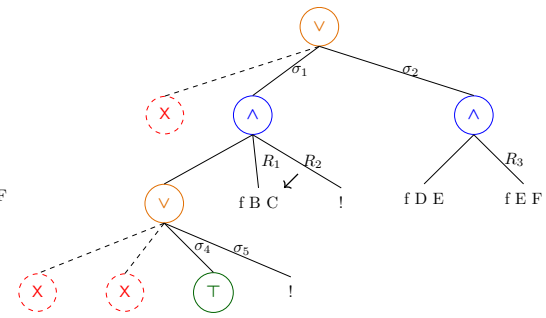


(e) step on Figure 6d.2

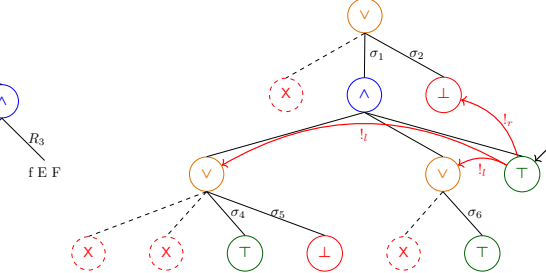
Figure 6 Some tree representations¹



(c) step on Figure 6b



(d.2) prune on Figure 6d.1



(f) step on Figure 6e

176 **next_tree** is the cut atom. Finally, **Expanded** accounts for the interpretation of predicate
 177 calls and non-superficial cuts.

178 The **step** procedure evaluates atoms. In particular, it leaves the tree unchanged when
 179 the tree is *success* or *failed*.

180 ► **Lemma 1** (*succ_step_iff*). $\forall p \ v \ \sigma \ t, \text{ success } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Success}, t).$

181 ► **Lemma 2** (*fail_step_iff*). $\forall p \ v \ \sigma \ t, \text{ failed } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Failed}, t).$

182 The **step** procedure expands **Todo** nodes by calling **backchain**, which returns a list l of
 183 alternatives. If l is empty, then the current call atom is replaced by **K0**. Otherwise, it is
 184 replaced by a right-skewed tree of **Or** nodes, where each leaf corresponds to a list of atoms
 185 $e \in l$. If e is empty, the leaf is **OK**; otherwise, it is a right-skewed tree of **And** nodes.

186 Figure 6 depicts the tree corresponding to the running example (Section 2.1) as it
 187 undergoes calls to **step**. We run query $q \ 0 \ R$ against the program P in Figure 1. Let c_0 ,
 188 c_1 , and c_2 be the children of the root. By construction, c_0 is the $\square$ tree, while c_1 and c_2
 189 correspond respectively to the premises of rules r_1 and r_2 in P . Note that c_1 (resp. c_2) is
 190 associated with substitution σ_1 (resp. σ_2), whose values are the same as in Section 2.1. Reset
 191 points are defines as follows: $R_1 := [\mathbf{f} \ Y \ Z, !]$, $R_2 := [!]$, and $R_3 := \mathbf{f} \ Y \ Z$. Intuitively, they
 192 record the lists of atoms used to generate the corresponding subtree. Other examples of
 193 **step** performing a tree expansion via backchain appear in Figures 6c–6e.

sottolineare le
 sostituzioni

194 3.2.1 The interpretation of a cut

195 The step corresponding to a cut performs two actions: (i) the cut node is replaced by **OK**; and
 196 (ii) the subtrees in the scope of **Cut** are pruned. More precisely, (ii) prunes the left siblings
 197 of the **Cut** node (in the same **And**-layer, denoted $!_l$) and prunes the right uncles of **Cut** (in
 198 the one-level-up **Or**-layer, denoted $!_r$).

199 An example of the impact of cut is shown in Figure 6f. The red arrows indicate the scope
 200 of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied to
 201 each pointed subtree. We note that the evaluation of a cut keeps the path leading to the
 202 cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under
 203 substitution σ_5 (which also succeeds) is replaced by **K0**. This amounts to discarding the third
 204 rule of predicate **r**.

205 3.3 The prune procedure

206 When **backchain** returns an empty list, **step** produces a *failed* tree and the computation
 207 must backtrack. The **prune** procedure is responsible for producing the new tree from which
 208 the computation can continue.

209 In Figure 6d.1 we encounter a failure because no rule applies to the atom $\mathbf{f} \ B \ C$ under
 210 substitution σ_3 , which maps **B** to 0. The **prune** procedure prunes the path leading to this
 211 failure and resets the current sub-query to its original shape. More precisely, it kills the **OK**
 212 node under σ_3 (since that branch has been fully explored and produced no solution), and
 213 then replaces the **K0** node with the information stored in the reset point R_1 . This restores
 214 the call $\mathbf{f} \ B \ C$, which can then be reconsidered under substitution σ_4 .

215 Furthermore, **prune** serves another important purpose. Its behavior depends on the
 216 boolean flag it receives as input: **prune** $\perp t$ recursively removes all failures (if any) from t ,
 217 whereas **prune** $\top t$ removes the first success in t . For example, the tree in Figure 6f contains
 218 a successful path that maps **R** to 2. Calling **prune** on this tree returns $\square$, since there are no
 219 alternatives in the tree after the success.

$$\begin{array}{c}
\frac{\text{success } t \quad \text{next_subst } \sigma \ t = \sigma' \quad \text{prune } \top \ t = t'}{\text{runT } p \ v \ \sigma \ t \cdot(\sigma', t') \perp v} \text{StopT} \\
\\
\frac{\text{incomplete } t \quad b' = (\text{tg} = \text{CutBrothers}) \vee b \quad \text{step } p \ v \ \sigma \ t = (v', \text{tg}, t') \quad \text{runT } p \ v' \ \sigma \ t' \ r \ b \ v''}{\text{runT } p \ v \ \sigma \ t \ r \ b' \ v''} \text{StepT} \\
\\
\frac{\text{failed } t \quad \text{prune } \perp \ t = \cdot t' \quad \text{runT } p \ v \ \sigma \ t' \ r \ n \ v'}{\text{runT } p \ v \ \sigma \ t \ r \ n \ v'} \text{BackT} \\
\\
\frac{\text{prune } \perp \ t = \square}{\text{runT } p \ v \ \sigma \ t \ \square \perp v} \text{FailT}
\end{array}$$

■ **Figure 7** Rule system for `runT`

Some interesting properties of `prune` are shown below; they illustrate how `prune` complements `step` with respect to failed, successful, and incomplete trees.

► **Lemma 3** (`incomplete_prune_id`). $\forall b \ t, \text{incomplete } t \rightarrow \text{prune } b \ t = \cdot t$.

► **Lemma 4** (`prune_Some`). $\forall b \ t \ t', \text{prune } b \ t = \cdot t' \rightarrow \text{failed } t' = \perp$.

► **Lemma 5** (`prune_None`). $\forall t, \text{prune } \perp \ t = \square \rightarrow \text{failed } t$.

3.4 The `runT` inductive

The inductive relation `runT` relates four inputs to three outputs. The inputs are a program, a set of variables, an initial substitution σ , and a tree t . The outputs are (i) an optional result r , (ii) a boolean b , and (iii) an updated set of variable names.

The main result r is $\square$ when the execution fails; otherwise when $r = \cdot(\sigma', t')$ the execution succeeded and produced the solution σ' ; t' represents the remaining, not-yet-explored alternatives. The boolean b records whether a superficial cut was evaluated during execution.

The `runT` predicate has four cases, depicted as inference rules in Figure 7. (*StopT*) If `next_tree` t is a success, then the substitution σ' is extracted from t (starting from σ) and the input tree is cleaned of its successful path. (*StepT*) If `next_tree` t is an atom, then `step` is invoked to evaluate t , and `runT` is called recursively on the updated tree. (*BackT*) If `next_tree` t is a failure, then `prune` is called to clear the failed path; if the resulting cleaned tree exists, `runT` is called recursively on it. Finally, (*FailT*) accounts for trees with no solutions.

The set of rules defining `runT` is deterministic.

► **Lemma 6** (`runT_det`). $\forall p \ v_0 \ \sigma \ t \ r \ r' \ b \ b' \ v \ v', \text{runT } p \ v_0 \ \sigma \ t \ r \ b \ v \rightarrow \text{runT } p \ v_0 \ \sigma \ t \ r' \ b' \ v' \rightarrow r' = r \wedge v' = v \wedge b' = b$.

The proof is by induction on the first `runT` derivation and is immediate, because the rules are selected based on `next_tree` t . In particular, the hypotheses `success` t , `incomplete` t , and `failed` t are mutually exclusive. Moreover, by ??, the hypothesis of *FailT* implies *failed* t ; hence *FailT* is exclusive with *StopT* and *StepT*, and it is also exclusive with *BackT* since they impose different requirements on the output of `prune`.

The boolean output of `runT` allows state interesting properties about the `Or` node in presence of cut. Here we only report a selection of the ones we proved.

► **Lemma 7** (`runSST_or`). $\forall p \ v \ v' \ \sigma \ \sigma' \ l \ l' \ \sigma_1 \ r, \text{runT } p \ v \ \sigma \ l \cdot(\sigma', \cdot l') \top v' \rightarrow$

$\text{runT } p \ v \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot (\sigma', \cdot (\cdot l' \vee KO_{\sigma_1})) \perp v'.$

248 ▶ **Lemma 8** (runNT_or). $\forall p \ v \ v' \ \sigma \ t \ \sigma_1 \ t', \text{ runT } p \ v \ \sigma \ t \ \square \top v' \rightarrow$
 $\text{runT } p \ v \ \sigma \ (\cdot t \vee t'_{\sigma_1}) \ \square \perp v'.$

249 ▶ **Lemma 9** (runNF_or). $\forall p \ v_0 \ v_1 \ v_2 \ \sigma \ l \ \sigma_1 \ \sigma_2 \ r \ r' \ b,$
 $\text{runT } p \ v_0 \ \sigma \ l \ \square \perp v_1 \rightarrow \text{runT } p \ v_1 \ \sigma_1 \ r \cdot (\sigma_2, r') \ b \ v_2 \rightarrow$
 $\text{let } sR := \text{omap } (\text{fun } x \Rightarrow \square \vee x_{\sigma_1}) \ r' \text{ in}$
 $\text{runT } p \ v_0 \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot (\sigma_2, sR) \perp v_2.$

250 Lemmas 7–9 correspond to the introduction rules for disjunction when the first disjunct performs
 251 a cut. If A executes a cut one *cannot* obtain $A \vee B$ out of B since the execution of A , be it in the
 252 end succesful or not, makes the success of B irrelevant (the cut discards it). In the first lemma B is
 253 forced to be KO , while in the latter the failure of A absorbs B . Depicted as rules:

$$\frac{A \quad (A \text{ cuts})}{A \vee \top} \quad \frac{\neg A \quad (A \text{ cuts})}{\neg(\neg A \vee B)} \quad \frac{\neg A \quad B \quad (A \text{ does not cut})}{A \vee B}$$

254 Finally, Lemma 10 is an elimination rule for disjunction.

255 ▶ **Lemma 10** (run_orSST). $\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ l \ l' \ r \ r',$
 $\text{runT } p \ v \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot (\sigma', \cdot (\cdot l' \vee r'_{\sigma_1})) \perp v' \rightarrow$
 $\exists b, \text{ runT } p \ v \ \sigma \ l \cdot (\sigma', \cdot l') \ b \ v' \wedge r' = \text{if } b \text{ then } KO \text{ else } r.$

256 ▶ **Lemma 11** (run_orSNT). $\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ l \ r \ r',$
 $\text{runT } p \ v \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot (\sigma', \cdot (\square \vee r'_{\sigma_1})) \perp v' \rightarrow$
 $(\exists b, \text{ runT } p \ v \ \sigma \ l \cdot (\sigma', \square) \ b \ v' \wedge \text{if } b \text{ then } r' = KO \text{ else } \text{prune } \perp \ r = \cdot r') \vee$
 $(\exists v_2 \ b, \text{ runT } p \ v \ \sigma \ l \ \square \perp v_2 \wedge \text{runT } p \ v_2 \ \sigma_1 \ r \cdot (\sigma', \cdot r') \ b \ v').$

257 From $A \vee B$, where A is not inhere (i.e. already explored) we cannot deduce A or B independently.
 258 We deduce only $\top \vee B'$ where B' provides no information if A performed a cut.

$$\frac{A \vee B \quad \frac{[A] \quad C \quad (A \text{ cuts})}{C}}{C} \quad \frac{A \vee B \quad \frac{[A] \quad C \quad [\neg A, B] \quad C \quad (A \text{ does not cut})}{C}}{C}$$

259 As anticipated in the intructions, these lemmas disprove the commutativity of disjunction.

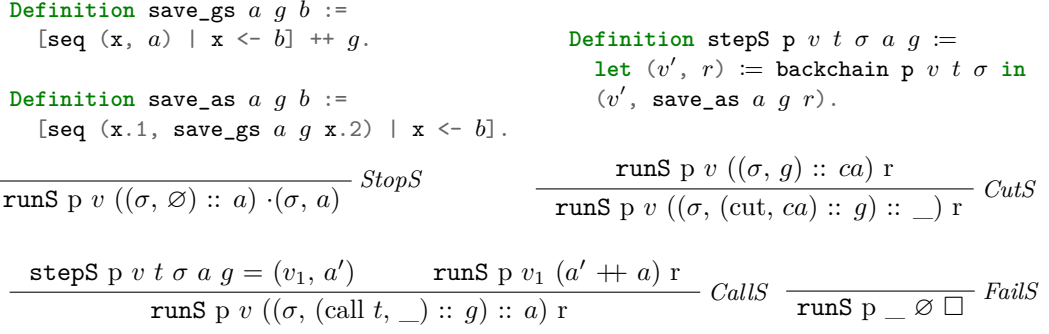
4 Stack-based semantics

261 The stack-based semantics we present is very close to what is implemented by the ELPI interpreter.
 262 This semantics is similar to the one proposed by Pusch in [20], which is in turn closely related to
 263 the semantics given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized this semantics in
 264 Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract
 265 Machine [26, 1].

266 The execution state is a stack of alternarives. Each alternative is a list of goals. Intuitively it
 267 amounts to a disjunctive normal form: each stack item is a self contained computations, coupling a
 268 substitution with all the goals to be solved.

269 Goals pair an atom with a list of the *cut-to* alternatives, making the two datatypes mutually
 270 recursive, but these alternatives have a very different role. They have to be understood as pointers
 271 to lower levels of the stack itself. This datum is used to implement the cut, i.e. to prune the stack
 272 of alternatives.

Inductive alts := nilA | consA of (Σ * goals) & alts
with goals := nilG | consG of (Atom * alts) & goals .



■ **Figure 8** Rule system for `runS`

273 The stack-based semantics is described by the `runS` inductive predicates.

274 **Inductive** `runS (p:  $\mathbb{P}$ ) :  $\mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop}$` (1)

275 The `runS` predicates relates a set of variables and a list of alternatives to optional result. $\square$

276 stands for failure, i.e., no solution, whereas $\cdot(\sigma, a)$ represent success with σ begin the solution and a

277 as the list of non-yet-explored alternatives.

278 The rule system for `runS` is given in Figure 8 and is made by 4 cases. We distinguish two main

279 cases. If we run out of alternatives we fail: (*FailS*) stops the execution and return $\square$. If the list of

280 alternatives is $(\sigma_0, h) :: a$, then we look at the list of goals h (under the substitution σ_0). If h is

281 empty, (*StopS*) stops the execution with a success σ_0 leaving a as the unexplored alternatives. If h

282 is not empty we look at its head goal (t, ca) where t is the atom and ca is its cut-to alternatives. If t

283 is a cut rule (*CutS*) continues to evaluate the tail of h under with alternatives ca . Finally, when

284 t is a call, rule (*CallS*) backchains: for each rule it pairs each premise with the current stack of

285 alternatives, and prepents the obtained goals to the tail of h . Note that if backchain returns the empty

286 list, (*CallS*) second premise amounts to exploring the next item in the current stack.

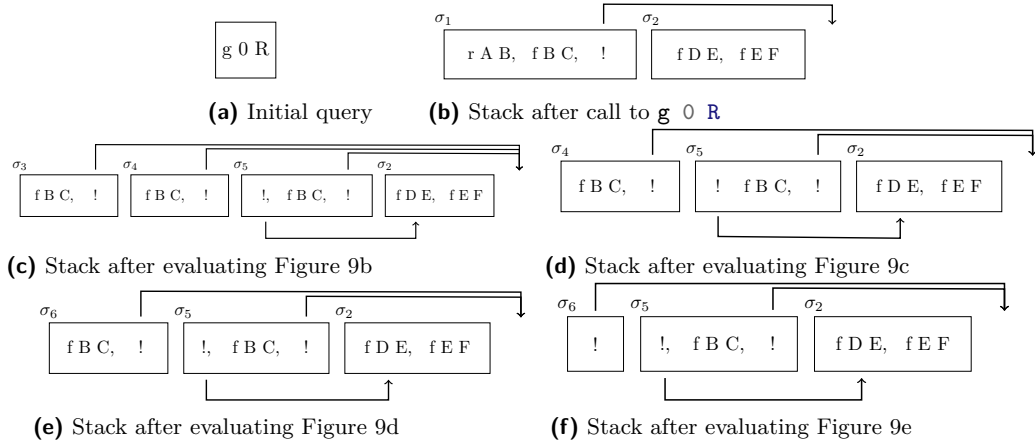
287 4.1 Example of execution

288 We propose an example of the execution of `runS` in Figure 9. The arrows in the images represent

289 the cut-to pointers. We have not drawn arrows coming out of call atoms since the interpreter

290 ignores the cut-alternatives in this case. The evolution of the stack in Figure 9 mirrors the example

291 in Section 2.1: we evaluate the first goal of the first alternative. During backchaining, we add a



■ **Figure 9** Example of execution

292 pointer to the remaining alternatives to each new cut operator. If a cut is evaluated, we replace the
 293 remaining alternatives with the cut-to alternatives. For example, in Figure 9f, evaluating the cut
 294 discards the second and third elements in the stack. Backtracking is performed by simply consuming
 295 the first alternative from the stack, as shown in Figure 9d.

296 4.2 Inadequacy of the stack-based semantics for formal reasoning

297 The lemmas that we presented in Section 3.4 are hard to express in the stack-based semantics, due
 298 to the lack of structure in the stack to delimit the scope of the cut operator. For example, consider a
 299 stack $a_0, a_1, \dots, a_n$. If a_0 succeeds but executes a cut, it is hard to relate the remaining alternatives
 300 $a_1, \dots, a_n$ to the alternatives $b_1, \dots, b_m$ that survive the cut, since the only simple invariant is that
 301 $b_1, \dots, b_m$ is a suffix of $a_1, \dots, a_n$. If $a_1, \dots, a_n$ were generated before the call whose execution
 302 generates the cut in a_0 , then they all stay; but if some were generated afterwards, they are discarded.
 303 Expressing this precisely would require attaching, at the very least, time-stamps (or depths) to goals,
 304 which amounts to a cumbersome encoding of a tree.

305 5 Equivalence of the two semantics

306 In order to relate the two semantics we have to relate the computations states, i.e., the And-Or tree
 307 and the stack of alternatives. We define a translation from trees to stacks, called `t2l`, but we do not
 308 explicitly provide the converse translation, an economy allowed by the proof technique we employ,
 309 inspired by [2].

310 Before describing the translation of trees to stacks (in Section 5.2) we need to define the notion
 311 of *valid tree*. The `tree` datatype indeed contains trees that cannot be generated by `runT` via `step` or
 312 `prune`, for example all the trees that do not correspond to a depth-first left-to-right tree construction
 313 strategy.

314 5.1 Valid trees

315 The class of valid trees is captured by the `valid_tree` function shown in Figure 10. While all base
 316 cases of `tree` are valid, we need to analyze the `Or` and `And` constructors more carefully.

317 For the `Or` constructor, we distinguish two cases depending on whether the left-hand side is
 318 present. If it is not, then the right-hand side must be a valid tree. Otherwise, the left-hand side
 319 must be a valid tree, and the right-hand side must either be the `K0` tree or be unexplored. The
 320 former case occurs when the right-hand side is cut away by the evaluation of a superficial cut in the
 321 left-hand side. In the latter case we use the `base_or` predicate to check that `B` is the right-skewed
 322 tree formed by a disjunction of conjunctions, generated after a successful backchain for a `call`.

323 For the `And` constructor, the left hand side is required to be a valid tree. The shape of the right
 324 hand side depends on whether the left hand side is a success. If it is not successful, then the right
 325 hand side must not be explored, i.e. `B` must be the right-skewed tree containing conjunctions of
 326 premise atoms. This is enforced using the `big_and` procedure, which generates a tree from the reset
 327 point B_0 (the same procedure used to transform each alternative output by `backchain` into the
 328 And-layer of the resulting tree). When the left hand side is successful, then the right hand side may
 329 have been explored, and consequently must be a valid tree.

330 5.2 From trees to stacks

331 The `t2l` recursive function translates a tree to a stack. For space constraints Figure 10 only gives
 332 the base cases, while we describe the recursive cases in the following text.

333 The function `t2l` takes as input the tree t_0 , a substitution σ_0 , and a list of choice points cp .
 334 These choice points represent alternatives that appear at the same level of the current tree, such as,
 335 for example, the right siblings of an `Or` node. The substitution is used to determine under which
 336 substitution the subtree being computed lives.

337 The `OK` node represents one successful alternative; therefore it is translated into a singleton
 338 list whose only element has an empty list of goals. The `K0` node represents failure, hence it is

```

Fixpoint valid_tree A :=
  match A with
  | Todo _ | OK | KO => T
  | Or □ _ B => valid_tree B
  | Or ·A _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

Fixpoint t2l t σ (cp: alts) : alts :=
  match t with
  | OK => [(σ, ∅)]
  | KO => ∅
  | TA a => [(σ, [(a, ∅)])]
  ...

```

■ Figure 10 Valid tree and Tree to list

mapped to the empty list of alternatives. Finally, atoms are mapped to a singleton list containing one alternative whose only goal is that atom, with an empty cut-to list. At this stage, the cut-to list cannot point to cp : atoms are not right-uncle subtrees, whereas cp represents alternatives that will actually be discarded if the atom turns out to be a cut. The correct cut-to list is therefore determined later, once the trees corresponding to sibling subtrees have been inspected.

The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for B . The alternatives of B are valid choice points for A and should be passed to the translation of A . The two lists of alternatives can then be concatenated. Moreover, every atom occurring one level below cp (i.e. in A or B) should add cp as its cut-to alternatives.

The translation of $A \wedge_{B_0} B$ starts with the translation of A . If the translation of A is an empty list we return the empty list of alternatives without even touching B nor B_0 , since an empty translation for A indicates failure and this in turn implies the failure of the entire conjunction. When the translation of A is a list $(\sigma_0, g) :: a$, the B node represents the continuation of the first alternative g , whereas B_0 is the continuation for each alternative in a .

5.2.1 Example of t2l

The translation of the trees in Figure 6 is given in Figure 9. The letters in the figures relate the trees to the corresponding letters in the stack figures. For example, the tree 6b is translated into 9b. We note that the two trees 6d.1 and 6d.2 are both translated into 9d, showing that $t2l$ is not injective; that is, there are different trees that map to the same stack. This means that there are transitions in runT that are not visible in runS .

As an example we focus on the call to $t2l$ on the tree 6c, and in particular its deepest **Or** node. This subtree is a disjunction with four children. The first is ignored, since it is $\square$. The success node below σ_3 has **f B C** and the cut operator as continuation, corresponding to the first cell in the stack in 9c. We note that the cut operator points outside the stack, since the scope of this cut eliminates its right uncle. The success node below σ_4 has R_1 as continuation, where R_1 is the list of goals **f B C, !**; this is translated into the second cell of the stack in 9c. Finally, the cut operator under σ_5 also has R_1 as continuation. It corresponds to the third alternative in Figure 9c. We can see that the first cut points to the translation of the subtree under σ_2 , since it is one **Or**-layer above its right uncles.

5.3 Equivalence proof

The tree-based semantics exposes more information than needed, hence we hide it behind existentials.

► **Definition 12** ($\text{runT}' p v \sigma t r$). $\exists b v', \text{runT } p v \sigma t r b v'$.

From Figures 6 and 9, we observe that, upon backtracking, runT may perform more steps than runS before returning a solution or a failure. These silent transitions must be skipped when proving the equivalence between the two semantics. Therefore, we prove the following lemma.

► **Lemma 13** (pruneF_t2l). $\forall t t' b, \text{valid_tree } t \rightarrow \text{failed } t \rightarrow \text{prune } b t = .t' \rightarrow$

$$\forall l \sigma, t2l \ t \ \sigma \ l = t2l \ t' \ \sigma \ l.$$

375 **Proof.** By induction on the tree t . ◀ dire di v

376 The first direction of the equivalence states that the execution of a valid tree is matched by
 377 the execution of the stack obtained by translating the tree. Moreover, the unexplored alternatives
 378 returned as a tree correspond to those returned as a stack.

379 ▶ **Theorem 14** (tree_to_elpi). $\forall p \ t \ \sigma \ r, \text{let } v := \text{vars_tree } t \ \backslash \backslash \text{vars_sigma } \sigma \text{ in}$

```

valid_tree t →
  runT' p v σ t r →
    let a := t2l t σ ∅ in
    let r' := if r is ·(σ', t) then ·(σ', t2l (odflt KO t) σ ∅)
              else □ in
    runS p v a r'.

```

380 **Proof.** We proceed by induction on the execution of runT' . There are four cases to consider. The
 381 first case arises from StopT ; it deals with successful trees. A successful tree must start with an empty
 382 list, and therefore we conclude using StopS . The case for StepT requires treating two subcases: step
 383 evaluates either a cut or a call. Accordingly, we apply CutS or CallS , followed by the induction
 384 hypothesis. The case for BackT is silent: it represents the non-injective behavior of t2l ; however,
 385 thanks to Lemma 13 and the induction hypothesis, the conclusion is proved. The last case arises
 386 from the derivation FailT . It is easily proved using FailS and the fact that if prune returns $\square$ when
 387 trying to erase failure in t , then t2l applied to t returns the empty list. ◀

388 The converse direction is stated following [2]: instead of using a function to translate a stack
 389 into a tree it states that any tree that translates to the given stack has an equivalent execution, i.e.
 390 gives the same solution and returns a unexplored tree that translates to the unexplored stack.

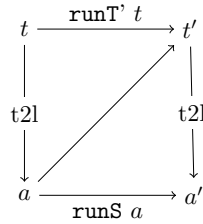
391 ▶ **Theorem 15** (elpi_to_tree). $\forall p \ v \ a \ r, \text{runS } p \ v \ a \ r \rightarrow$

```

∀ σ t, valid_tree t → t2l t σ ∅ = a →
  if r is ·(σ', a') then
    ∃ t', runT' p v σ t ·(σ', t') ∧ t2l (odflt KO t') σ ∅ = a'
  else runT' p v σ t □.

```

392 **Proof.** The proof is based on the idea explained in [2, Section 3.5.1] and depicted in the figure on the right: we have a state t whose translation
 is a , we proceed by induction on the execution of runS . This gives us not
 only the output a' but that hypothesis that it exists a tree t' such that
 its translation to the stack state is a' . This proof strategy is indirectly
 providing a kind of t2l function by combining the execution of the runS
 and t2l . ◀



393 Stating the converse direction in a symmetric way would require a function t2l transforming a
 394 stack a into a tree t . Writing this function is far from trivial, since the structure of the tree is lost by
 395 save_as and runT that intuitively keep the state as a big disjunction of conjunctions by distributing
 396 conjunctions over disjunctions. Moreover, one would need to define a notion of valid stack that is
 397 equally intricate.

398 The constructive nature of the logic of Rocq tells us that the proof above provides an effective
 399 function translating stacks into trees. However, that function also uses an execution trace of the
 400 stack-based semantics, from which it recovers the missing tree structure.

6 Case study: determinacy analysis

402 An interesting application of the tree semantics is determinacy checking, introduced in version 3 of
 403 the ELPI language [11]. The checker takes as input the signatures of the predicates declared by the

user and verifies that the rules of a deterministic predicate generate at most one solution; we call this particular kind of predicate a function. Thanks to this static verification, the user can better control unexpected backtracking statically.

A predicate behaves deterministically if two conditions are satisfied: *mutual exclusion* guarantees that at most one rule can be successfully applied to any given query, whereas *rule checking* ensures that each rule does not create choice points, for example by calling non-deterministic predicates.

Determinacy checking is strongly related to the modes of predicates: we differentiate between input and output arguments. A deterministic call relates to its inputs only one output. In the literature, we find several works about determinacy checking, namely [21, 16, 7]. These works need a mode checker statically ensuring that in a query with ground inputs, each recursive call is performed with ground inputs.

In our mechanization, and in particular in the ELPI language, we adopt a special unification routine for input argument, called **matching** [1, 27]. Similarly to **unify**, **matching** takes the query-term t_1 and the pattern t_2 and returns an optional substitution σ' that does not assign any variable in the query.

The function **matching** has the following property:

► **Axiom 1** (**matching_disj**). $\forall t_1 t_2 \sigma \sigma', [disjoint_vars_tm\ t_1 \ \&\ \text{domf}\ \sigma] \rightarrow disjoint_tm\ t_1\ t_2 \rightarrow matching\ t_1\ t_2\ \sigma = \cdot\sigma' \rightarrow \exists e, \text{domf}\ \sigma' = \text{domf}\ \sigma \setminus e \wedge e \leq vars_tm\ t_2.$

If the two terms are disjoint and the variables in the support of σ are disjoint from the variables in t_1 , then, if matching succeeds, only the variables in t_2 are assigned; that is, $t_1 \equiv \sigma' t_2$, and the variables in t_1 are not assigned in σ' .

The **backchain** function uses **matching** or **unify** on the arguments q depending if the argument is in *input* or *output* mode. We assume two important properties of our unification routines.

► **Axiom 2** (**unif_trans**). $\forall t_1 t_2 t_3 \sigma, unify\ t_1\ t_2\ \sigma \rightarrow unify\ t_2\ t_3\ \sigma \rightarrow unify\ t_1\ t_3\ \sigma.$

► **Axiom 3** (**match_unif**). $\forall t_1 t_2 \sigma, matching\ t_1\ t_2\ \sigma \rightarrow unify\ t_1\ t_2\ \sigma.$

In Figure 1, the rules of the program are equipped with three signatures, one per predicate. Besides the arity of each predicate, a signature specifies which arguments are inputs and which are outputs, their types, and whether the predicate is deterministic. They indicate that **f** and **g** are expected to be functions, as specified by the **func** keyword, whereas **r** is a relation, as indicated by the **pred** keyword. The **->** symbol separates inputs from outputs; therefore, all three predicates are expected to take an **int** as input and return an **int** as output.

The signatures of the program are stored in the **sig** field of the program and are encoded as described in [11]; that is, similarly to the code in Figure 1, we use arrows for term application, data types to encode base types such as *int*, and two dedicated symbols to distinguish deterministic from non-deterministic predicates.

A program execution behaves deterministically if, given a program, a substitution, and a tree to the **runT** inductive, the execution either fails or, if it succeeds, the tree of alternatives is $\square$.

► **Definition 16** (**is_det** $p\ \sigma\ v\ t$). $\forall r, runT'\ p\ v\ \sigma\ t\ r \rightarrow r = \square \vee \exists \sigma, r = \cdot(\sigma, \square).$

The theorem we want to prove is the following one:

► **Theorem 17** (**det_check_call**). $\forall p\ \sigma\ t\ v, check_IP\ p \rightarrow tm_is_det\ p.(sig)\ t \rightarrow is_det\ p\ \sigma\ v\ (Todo\ (call\ t)).$

Here, **check_{IP}** denotes the function that checks the program with respect to mutual exclusion and rule checking, and **tm_is_det** denotes the function that tests whether the head of the term refers to a rigid constant with a deterministic type.

The proof of this lemma proceeds by induction on the program execution. However, without further work, the induction hypothesis is too weak since it talks about **Todo** trees (i.e. atoms), whereas we need it on any tree.

To address this issue, we introduce the notion of “deterministic trees” (**det_tree**), show that a tree satisfying this condition enjoys the desired properties, and then prove the following more general theorem:

452 ► **Lemma 18** (`det_check_tree`). $\forall \sigma \ v \ p \ t, \text{check_}\mathbb{P} \ p \rightarrow \text{det_tree } p.(\text{sig}) \ t \rightarrow \text{is_det } p \ \sigma \ v \ t.$

453 Our target theorem `det_check_call` is a corollary of `det_check_tree`.

454 7 Related work

455 We studied a PROLOG semantics in which input arguments are *matched*, rather than unified, against
 456 rule heads. This choice agrees with the ELPI interpreter, which is in turn inspired by B-Prolog’s
 457 “matching-clauses” [1, 27]. The two semantics we presented (and proved equivalent) also apply to
 458 standard PROLOG if one forces all predicate signatures to have only output arguments.

459 For determinacy analysis, matching input arguments affects the development only as follows:
 460 Axiom 1 does not require the query q to be ground. Adapting our results to standard PROLOG
 461 therefore requires the insertion of two additional hypotheses in Theorem 17. First, one has to require
 462 that t has ground inputs. Second, that the program p is well-moded. Well-moded predicates produce
 463 ground outputs when given ground inputs; hence, starting from an initial query with ground inputs,
 464 well-moded programs behave exactly as if input arguments were matched. Therefore, the proofs in
 465 Section 6 still apply.

466 The only mechanization of Prolog’s semantics we could find is the one by Pusch in ISA-
 467 BELLE/HOL [20] that is based on the model of Debray and Mishra [6, Sec. 4.3]. Their work
 468 starts from that semantics and refines it by introducing some optimizations usually found in the
 469 Warren abstract machine [26, 1]. They do not use the semantics to prove properties on the execution
 470 of programs as we do in our use case, hence they do not face the difficulty described in Section 4
 471 that pushed us to develop a more explicit semantics (with respect to cut). In a sense, our work is
 472 complementary, showing the stack-based semantics equivalent to a more optimized one.

473 Another relevant work is by King [15], but we could not find their Rocq source code. Their
 474 semantics is denotational and cannot easily cover all the features our paper covers, but it would
 475 have been sufficient for this first step.

476 The proof technique we use to relate the two semantics is inspired by [2], and we thank Y. Bertot
 477 for insightful discussions on the topic. In [2], Bertot mechanizes the correctness of a compiler. He
 478 relates the semantics of an imperative language to that of its compiled assembly code and proves
 479 their equivalence. His approach uses the compiler as the translation function from one language
 480 to the other, but does not introduce a decompiler, just as we do not define a function from stacks
 481 to trees. To show that the assembly code behaves like the imperative language, he proceeds by
 482 induction on the execution of the assembly program. We adopt the same strategy in the proof of
 483 Theorem 15.

484 8 Conclusion

485 This paper describes two operational semantics for Prolog with cut. The first semantics is based on
 486 And–Or trees: it is well suited to graphically illustrating the scope of cut, and to proving lemmas
 487 about the impact of its evaluation when reasoning about disjunctions. The second semantics is
 488 based on a stack of alternatives: it is computationally more efficient, but it loses the structural
 489 expressiveness of the tree-based semantics. Thanks to a dedicated function, we translate trees into
 490 stacks and prove the two semantics equivalent. This stack-based semantics corresponds to the
 491 first-order fragment of ELPI.

492 Using the tree-based semantics, we mechanize the first-order part of the determinacy checker that
 493 has been available in ELPI since version 3.0, and we prove that a call to a deterministic predicate
 494 returns at most one solution.

495 To fully model the ELPI language, we would need to account for higher-order variables, i.e.
 496 variables representing predicates. This extension is ongoing and largely orthogonal to the present
 497 paper, since cut and higher-order features do not interact in subtle ways.

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